CodeShowOff / General

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

General / README.md in main

Cancel changes Commit changes...

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A **monotonic stack** is a type of stack that maintains its elements in either **increasing** or **decreasing** order. It is commonly used in problems involving **next greater element, next smaller element**, and similar range queries.

Types of Monotonic Stacks

1. **Monotonically Increasing Stack**
   - Maintains elements in **increasing order** (smallest on top).
   - Before pushing a new element, it pops all **larger** elements.
   - Useful for **finding the next smaller element**.
2. **Monotonically Decreasing Stack**
   - Maintains elements in **decreasing order** (largest on top).
   - Before pushing a new element, it pops all **smaller** elements.
   - Useful for **finding the next greater element**.

Example Problem: Next Greater Element

Given an array, find the next greater element for each element.
Example:
Input: `[2, 1, 4, 3]`
Output: `[4, 4, -1, -1]`

Solution Using Monotonic Stack

1. Traverse the array from **right to left**.
2. Maintain a **monotonic decreasing stack** (stores elements in descending order).
3. For each element:
   - Pop elements from the stack **until you find a greater element**.
   - If the stack is empty, assign `-1` (no greater element).
   - Push the current element onto the stack.

Why Use a Monotonic Stack?

- **Efficient:** Processes each element **only once → O(n) time complexity**.
- **Simple:** Avoids **nested loops** that would result in $O(n^2)$ complexity.
- **Versatile:** Used in problems like **stock span, temperature forecasting, histogram area, etc.**

Let me know if you need more examples! 🚀